

EFFECT OF MECHANOCHEMISTRY ON ORGANIC MATTER: IMPLICATIONS FOR IMPACT GARDENING ON ASTEROID SURFACES

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Introduction: Organic matter in carbonaceous chondrites likely derives from precursors synthesized during pre-accretion events in the protosolar nebula or parent molecular cloud [1]. Although hydrothermal and radiative processes may later have modified this organic inventory, the role of shock-induced impacts remains poorly understood. Indeed, studies of carbon-rich asteroids have evidenced a “rubble-pile” structure, indicating that these bodies formed and evolved through successive impact and re-accretion events [2]. Moreover, asteroid surfaces are made of loose regolith, indicating that these bodies have undergone impact processes, from micrometeoroid bombardment to larger collision events, throughout their evolution [3, 4]. Hexamethylenetetramine (HMT) and its derivatives, ubiquitous in UV-irradiated laboratory analogues of interstellar ices [5], have been identified in extracts of the Murchison meteorite [6]. Under asteroidal hydrothermal conditions, HMT decomposes into ammonia and formaldehyde, key intermediates in the synthesis of soluble organics in extraterrestrial materials [7].

Methodology: Here, we investigate the solid-state mechanochemical (*i.e.*, 25 Hz, 90 min) reactivity of HMT with Na-rich montmorillonite (MMT) as a model system to assess how impact-driven processes influence the evolution of extraterrestrial organic matter. The “cyclic mechanical input” delivered by laboratory ball-milling, which produces high-energy impacts, and grinding, which applies a lower-energy mechanical input, serves as an analogue for shock processes on asteroid surfaces, including repeated impacts by micrometeorites and interplanetary dust particles (IDPs)—a process known as space gardening.

Results: HMT is highly reactive under mechanical stress, particularly when the clay mineral structure is disrupted under milling conditions. This regime induces the formation of a diverse suite of compounds, including HMT-related species (*e.g.*, HMT-CH₃, HMT-OH), pyrazine, and triazinane/triazine derivatives. Notably, the incorporation of oxygen in *N*-rich species happens even in the absence of liquid water. Conversely, under grinding conditions, the clay mineral structure remains intact, promoting the solid-state intercalation of HMT-like molecules (HMT, HMT-CH₃, and a triazinane derivative) into the clay interlayer space, effectively protecting them from further chemical transformation.

Conclusions: Our findings demonstrate that impact events are key drivers of both the transformation and preservation of extraterrestrial organics, offering new insight into how such processes may have shaped the organic inventory of small bodies and influenced the exogenous delivery of possible prebiotic molecules to planetary surfaces.

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